

Predictive Regression Analysis on Profitability

1. Introduction

The purpose of this project was to analyze the factors affecting company profit and build a predictive regression model to estimate future profit outcomes. Before selecting a prediction goal, a basic exploratory data analysis (EDA) was conducted to understand relationships between business variables such as sales, quantity, discount, category, region, and shipping mode.

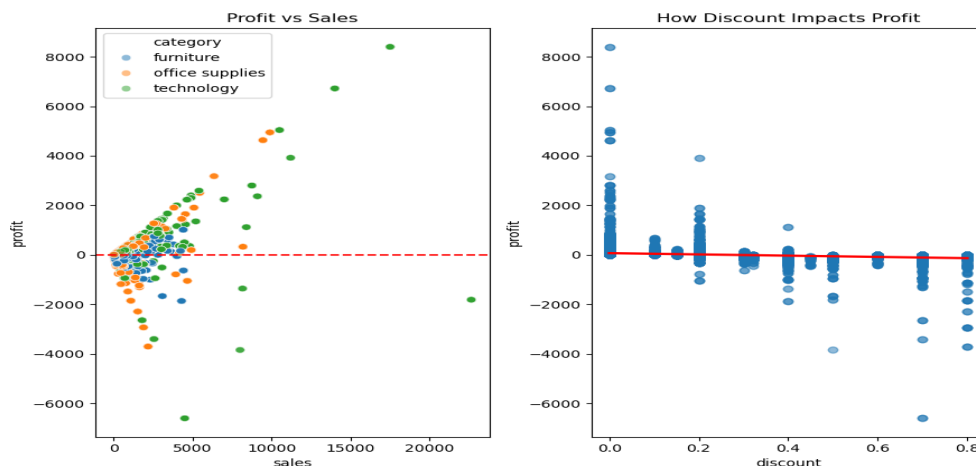
During the correlation analysis, it was observed that discount had the strongest negative relationship with profit compared to other features. This raised an important business question:

“Is discount actually hurting company profitability?”

To answer this question, a regression-based machine learning model was developed to predict profit and analyze the true impact of discounts on business performance.

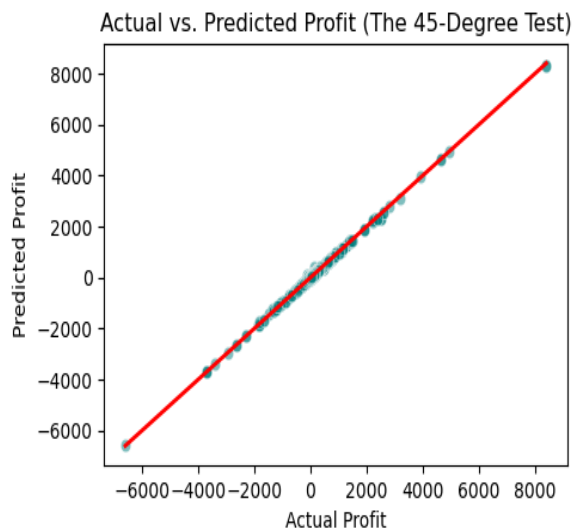
2. Business Objective

- To predict profit using historical sales data
- To identify whether discount is the major reason for profit loss
- To understand how different business components influence profitability
- To provide business recommendations for improving profit



3. Model Selection and Performance

We selected **features: 'sales', 'quantity', 'discount', 'category', 'region', 'ship_mode'** and **targeted 'profit'**. The model was trained using a Random Forest Regressor as it works well with mixed feature types. The model achieved extremely high predictive performance, showing that the selected business features strongly explain profit behavior.

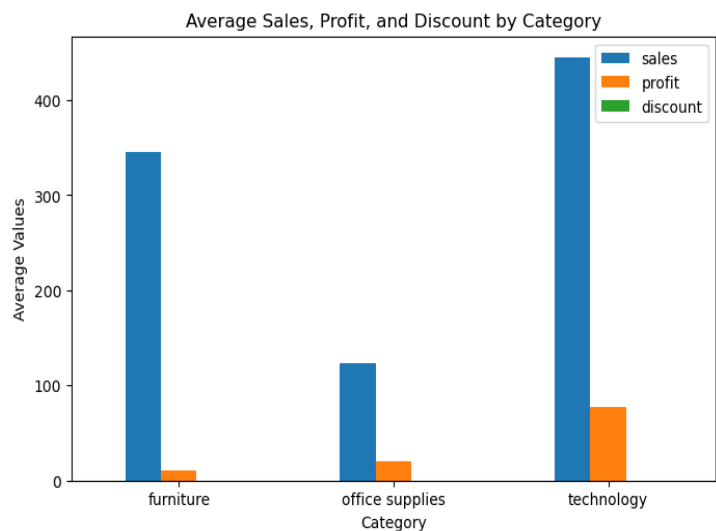


Accuracy: 0.9989

Error: 1.4735

To delve deeper into how discounts affect profit, we analyzed performance by category:

4. Category-Level Business Insights



Technology: High sales, high profit. Business strategy appears healthy.

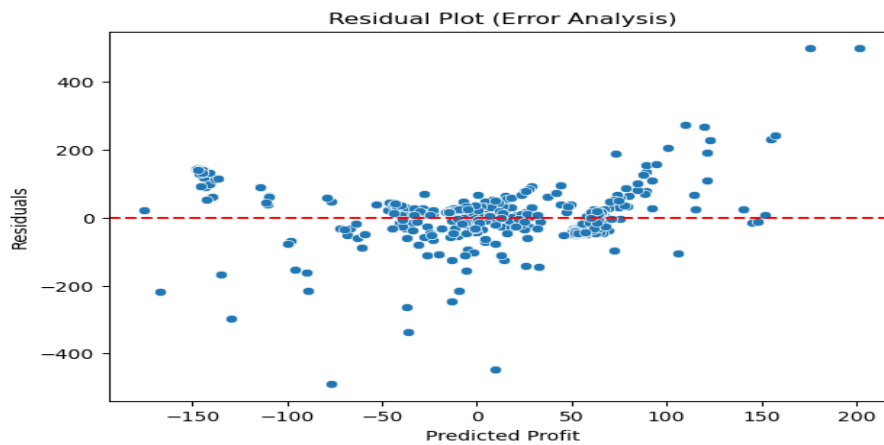
Office Supplies: Low sales, low profit. Requires demand improvement strategies.

Furniture: High sales but low profit. Profitability issues exist despite strong demand.

This raised the question: Is discount dependency reducing Furniture profitability?

So Regression Predictive analysis was performed specifically for the Furniture category:

5. Discount Impact Analysis (Furniture)



R2 Score: 0.3603 MAE: 50.49

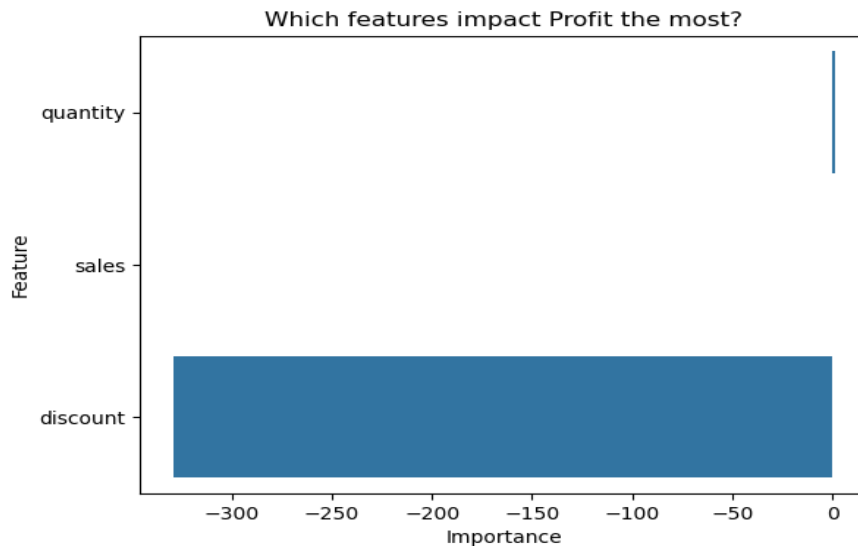
6. Discount Simulation Analysis:-

A simulation was performed on multiple Furniture samples to validate the impact of increasing discount levels. As discount increased, predicted profit consistently decreased, eventually turning negative.

Low Sales :-

44.46	2	0.0	52.09
44.46	2	0.1	19.18
44.46	2	0.2	-13.73
44.46	2	0.3	-46.64
44.46	2	0.4	-79.55

19.54	2	0.0	51.03
19.54	2	0.1	18.18
19.54	2	0.2	-14.73
19.54	2	0.3	-47.64
19.54	2	0.4	-80.55



7. Final Business Findings

- High discounts are the biggest contributor to profit loss.
- Even high sales transactions can become unprofitable due to excessive discounting.
- Furniture category is heavily dependent on discount-based sales.
- The ML model strongly confirmed the inverse relationship between discount and profit.

8. Final Recommendations

1. Discount Optimization: Limit excessive discounts. Recommendation: Keep discounts below 20% whenever possible.

2. Customer Segmentation (RFM): Identify high-value customers and avoid unnecessary discounting for loyal segments.

3. Market Basket / Combo Strategy: Encourage bundled purchases instead of large individual discounts.

4. Category-Level Strategy:

- Technology: Maintain current healthy strategy.
- Office Supplies: Focus on increasing demand and sales campaigns.
- Furniture: Reduce discount dependency and redesign pricing strategy.

9. Conclusion

This project successfully analyzed the factors affecting profitability using predictive regression analysis.

The Random Forest Regressor model achieved highly accurate predictions and confirmed that excessive discounts are significantly reducing company profit.

Although discounts help attract customers and increase sales volume, uncontrolled discounting can turn profitable transactions into losses.

The analysis also demonstrated that profit behavior is influenced by multiple business components, making it important to balance sales growth with profitability strategies.

The final recommendations provide actionable business solutions for improving long-term profit performance while maintaining customer demand.